

Multimedia Metadata

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Lecture Overview: Multimedia Metadata

- ▶ MM data vs the real world: expression gap, semantic gap, data diversity
- ▶ Metadata fundamentals: types, layers, extraction (manual and automatic)
- ▶ Metadata by media type: text, image, audio, video, and integrated AV data
- ▶ Panofsky's three levels: description, identification, interpretation
- ▶ Metadata management: architectures, standards, controlled vocabularies
- ▶ Goal: design metadata that bridges human meaning and machine-processable description



Multimedia data and Real world

- ▶ Difference in “expression”
 - ▶ Ability to express the perception and representation of MM data about an entity
 - ▶ One entity can always be presented by multiple multimedia data (with different distances, views, brightness,...)
 - ▶ Multiple different entities can also be described by the same dataset
 - ▶ The difference between the “expression” of a machine and the senses of a person

Multimedia data and Real world

- ▶ Difference in “expression”



Multimedia data and Real world

- ▶ Difference in “expression”



"My wife and my mother-in-law" was published in 1915 by cartoonist W.E. Hill.

Multimedia data and Real world

▶ Semantic Difference

- ▶ Information about entities is encrypted and stored in different data formats,
- ▶ Processed information is often compromised, or difficult to convey semantics easily.
- ▶ There is always a gap between the true semantics that humans understand about the entity and the information we store in computers.

The Semantic Gap in Multimedia Systems

- ▶ Low-level features: pixels, MFCC coefficients, color histograms — machine-computable
- ▶ High-level semantics: 'a wedding ceremony', 'sad melody', 'historical portrait' — human interpretation
- ▶ Semantic gap: distance between extracted features and user-intended meaning
- ▶ Same entity → many MM representations (photo, 3D scan, description text) with different expressive power
- ▶ Same dataset → may describe multiple entities (ambiguous image, shared audio sample)
- ▶ Metadata role: annotate media with structured descriptions to narrow the semantic gap



Multimedia data and Real world

▶ Data diversity

- ▶ State-of-the-art technology allows the creation and storage of a wide range of multimedia data
- ▶ Multimedia devices can be used easily
- ▶ The information contained in each data is rapidly decreasing in "value"
- ▶ A reasonable storage mechanism is needed for digital data

Multimedia data and Real world

- ▶ Reality at museums: facing all the stated problems
 - ▶ An artwork / entity can be displayed/stored in various forms of data.
 - ▶ Any form of data cannot fully express the semantics of a work/entity.
 - ▶ Multiple data stored about the same thing diminishes the expression "value" of each data.

Multimedia data and Real world

- ▶ New issues arise

- ▶ The temporal structure of MM data requires new techniques (vocabulary) of expression.
- ▶ Video or audio used to express emotions/psychology is often structurally far different from presenting in text data. There needs to be a separate way of expressing these data.
- ▶ Example: We don't usually seek information in music data but often look for entertainment or spiritual values.

Why Metadata Matters for Multimedia Databases

- ▶ Discovery: enable search and browse without processing raw BLOBs every query
- ▶ Management: access control, rights, provenance, version history (administrative metadata)
- ▶ Preservation: format, codec, checksum, migration history for long-term archives
- ▶ Interoperability: shared schemas (Dublin Core, MPEG-7) allow exchange across systems
- ▶ Integration: link text, image, audio, video describing the same entity (cross-media metadata)
- ▶ As data volume grows, metadata value increases — 'data about data' becomes the primary access path



Multimedia metadata

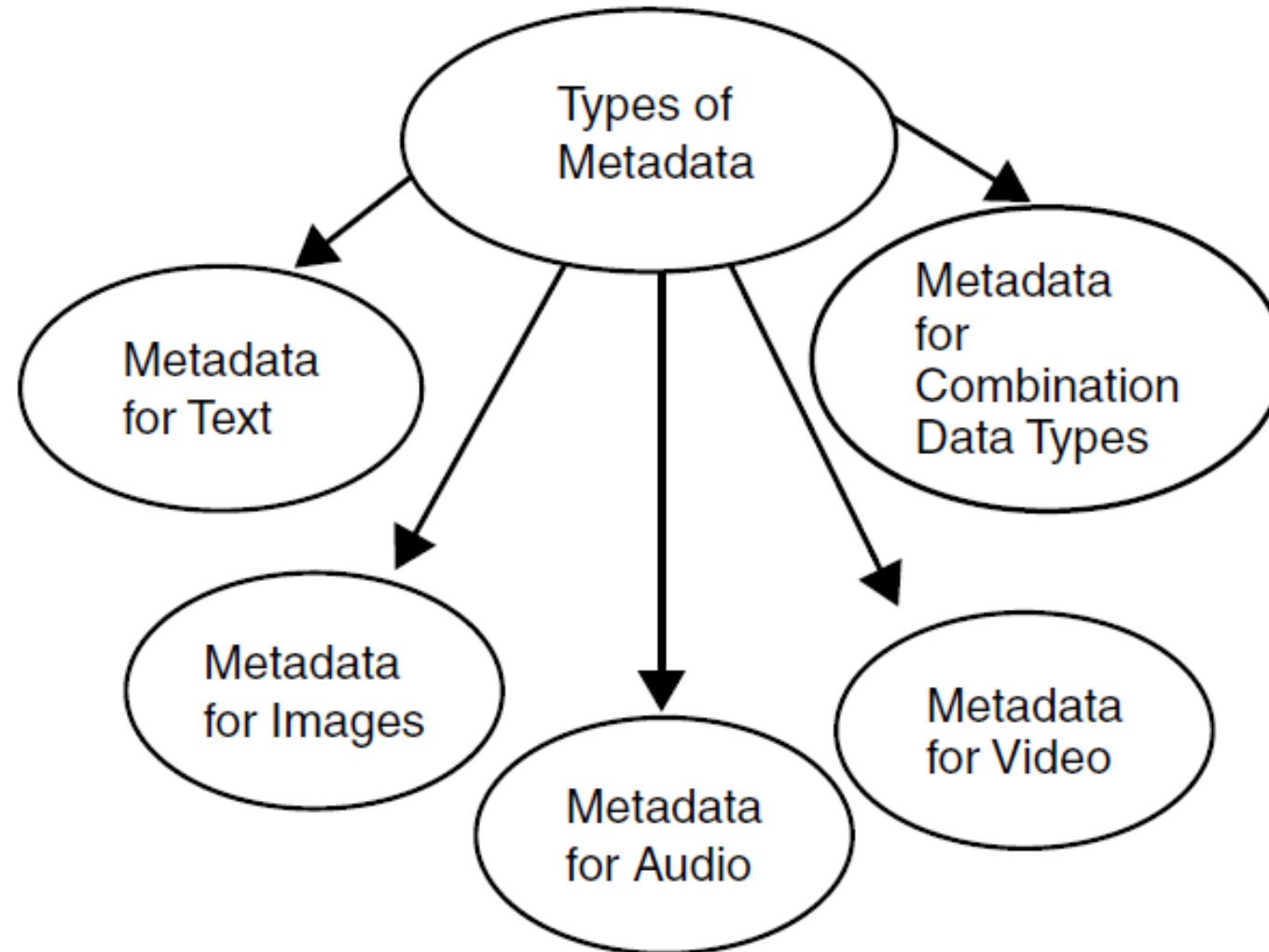
- ▶ Structured data based on classification concepts.
- ▶ Data described by layer
- ▶ Information about the entities contained in the data

Taxonomy of Metadata Types

- ▶ Descriptive metadata: title, creator, subject, description — WHAT the resource is about
- ▶ Structural metadata: chapters, scenes, segments, page order — HOW parts relate internally
- ▶ Administrative metadata: rights, license, creation date, preservation policy — HOW to manage it
- ▶ Technical metadata: format, resolution, codec, bitrate, file size — HOW it is encoded
- ▶ Behavioral/usage metadata (optional): access logs, annotations, user tags
- ▶ Layered model: technical → descriptive → semantic/interpretive (maps to Panofsky levels)



Multimedia metadata

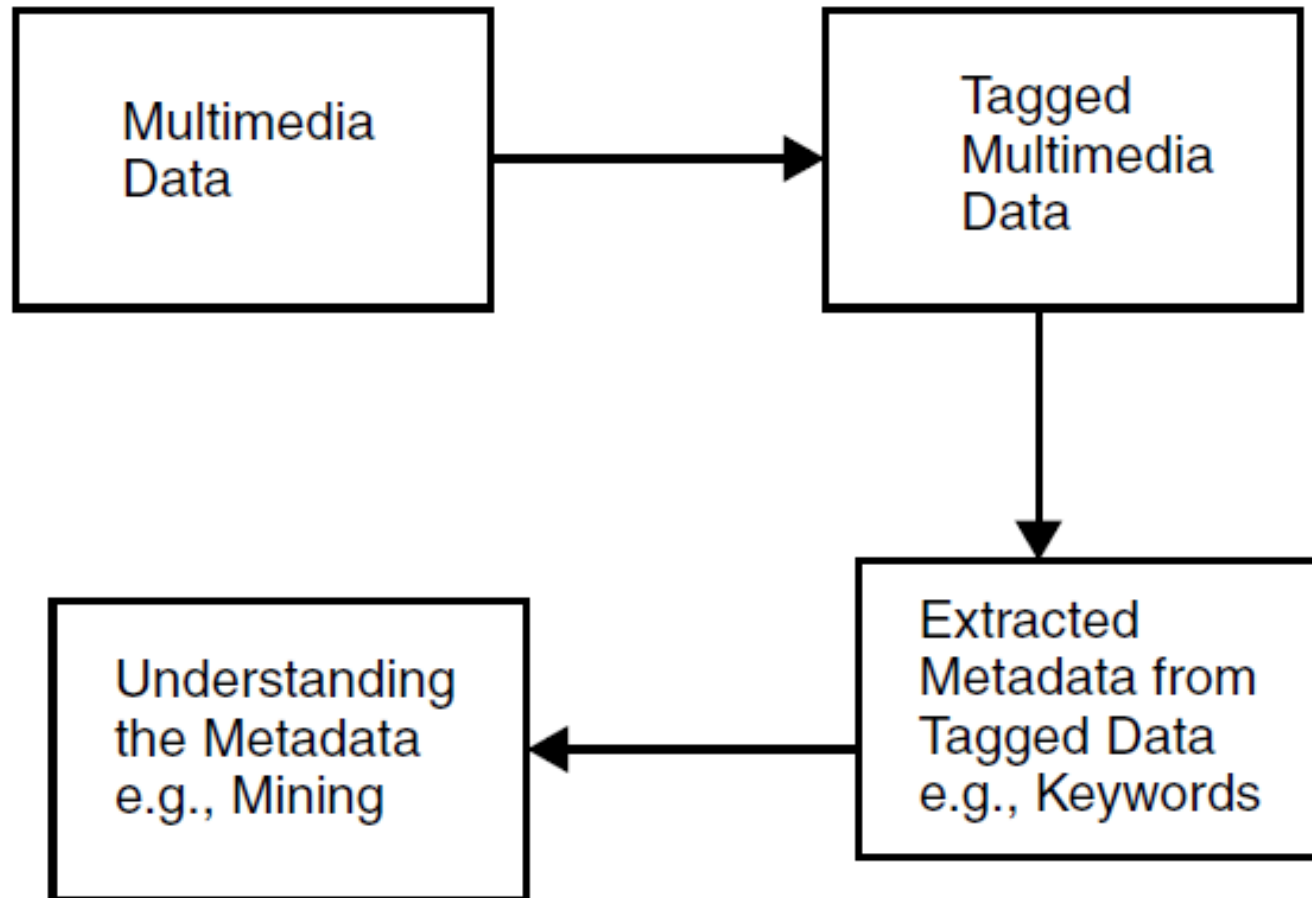


Metadata Standards and Layered Schemas

- ▶ Dublin Core (DC): 15 core elements — simple cross-domain descriptive metadata (Title, Creator, Subject...)
- ▶ MPEG-7: rich multimedia content description — visual, audio, multimedia description schemes (MDS)
- ▶ EXIF / IPTC: embedded image metadata (camera settings, caption, keywords, GPS)
- ▶ ID3 / Vorbis comments: embedded audio tags (artist, album, genre, track number)
- ▶ Layer principle: combine lightweight catalog record (DC) with media-specific deep metadata (MPEG-7)

Multimedia metadata

- ▶ Metadata extraction from a multimedia data.



Metadata Extraction: Manual vs Automatic

- ▶ Manual annotation: expert catalogers, curators — high semantic quality, slow and costly
- ▶ Automatic extraction:
 - ▶ Text: NLP for keywords, entities, summaries
 - ▶ Image: EXIF parsing, CBIR features (color, texture, shape), captioning (AI)
 - ▶ Audio: speech recognition, music genre/tempo detection, speaker ID
 - ▶ Video: shot/scene detection, keyframe extraction, object tracking
- ▶ Hybrid pipeline: auto-extract technical + low-level features; human validates interpretive metadata



Multimedia metadata

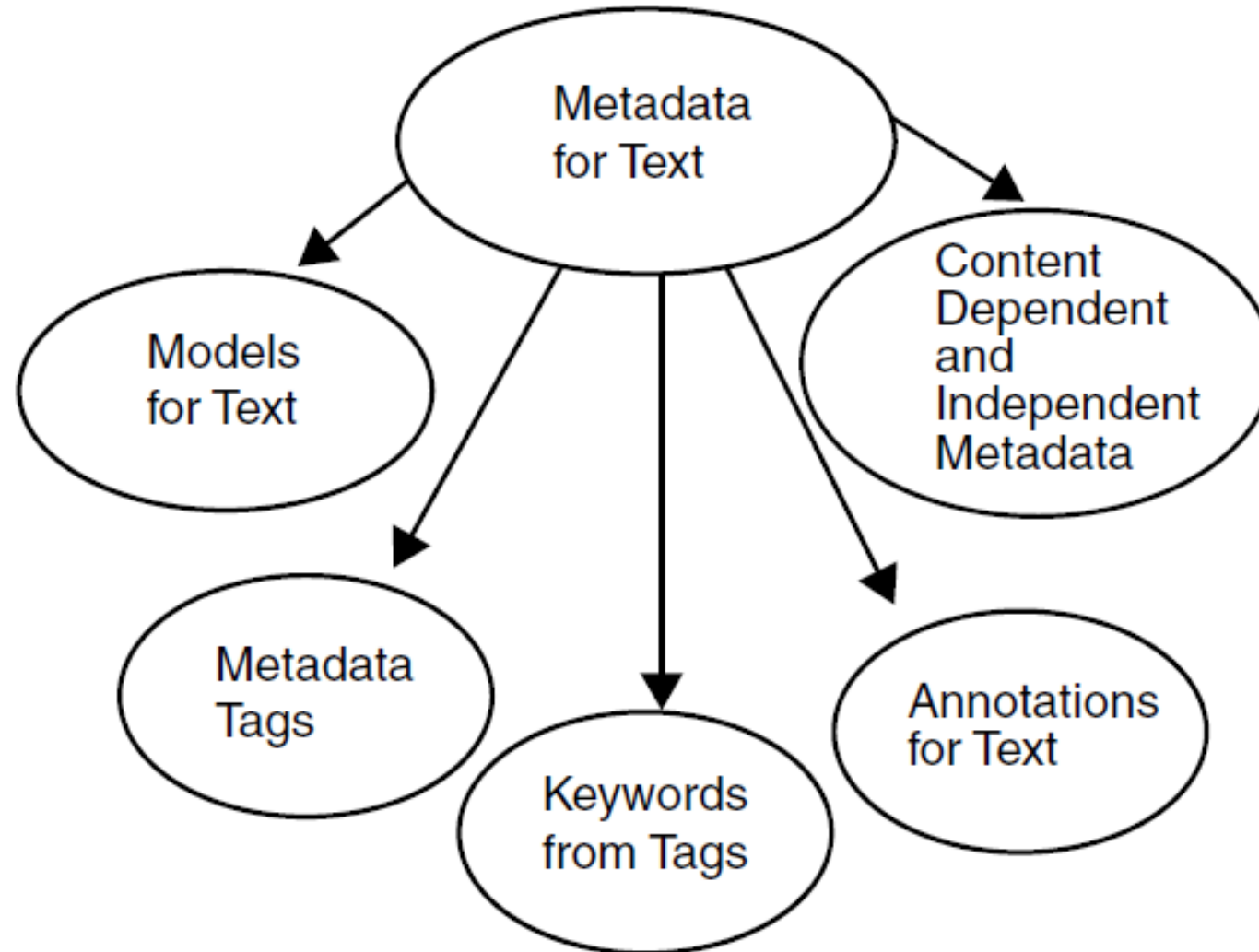
- ▶ Metadata for Text (books, newspapers, essays,...):
 - ▶ Includes text type, page number, word count, chapter number, paragraph number, text format, plot information, keywords,...
 - ▶ Text metadata modeling: relational data models, object-oriented data models, indexing and retrieval systems (IR Systems).
 - ▶ Text metadata can be represented by a catalog of annotations.

Text Metadata — Standards and Modeling

- ▶ Common fields: document type, language, page count, chapter/section, word count, format (PDF/HTML)
- ▶ Plot/narrative metadata: characters, events, themes (literary catalogs, FRBR model)
- ▶ Modeling options: relational tables (CatalogRecord), OO classes (Document hierarchy), IR index terms
- ▶ Annotation catalog: ordered list of labeled attributes attached to a text object
- ▶ Standards: TEI (Text Encoding Initiative), MODS (Metadata Object Description Schema), MARC
- ▶ Link to Lecture 8: full-text index (inverted file) complements bibliographic metadata catalog



Multimedia metadata



Example — Text Metadata Catalog Record

- ▶ Document: 'Introduction to Multimedia Databases'
 - ▶ Title: Introduction to Multimedia Databases
 - ▶ Author: Nguyen, H.D. | Language: en | Format: PDF | Pages: 42
 - ▶ Keywords: {multimedia, database, metadata, indexing}
 - ▶ Chapter: 1 | Section: 1.2 Metadata concepts
 - ▶ Structural: TOC → [1.1 Overview, 1.2 Metadata, 1.3 Applications]
 - ▶ Rights: CC-BY-NC | Created: 2014-09-01 | LastModified: 2026-09-15

Multimedia metadata

▶ Image metadata

- ▶ May include text describing the content of the images. For example: "In the picture is a smiling young woman, with a rose in her hand, in the distance is a ripe plum-colored car parked next to a yellow house with a red roof".
- ▶ Metadata stored using a relational data model that describes various properties of an image.
- ▶ Metadata includes a set of annotations
- ▶ Image metadata includes photos.
- ▶ Content-based metadata
- ▶ Metadata includes all photo features

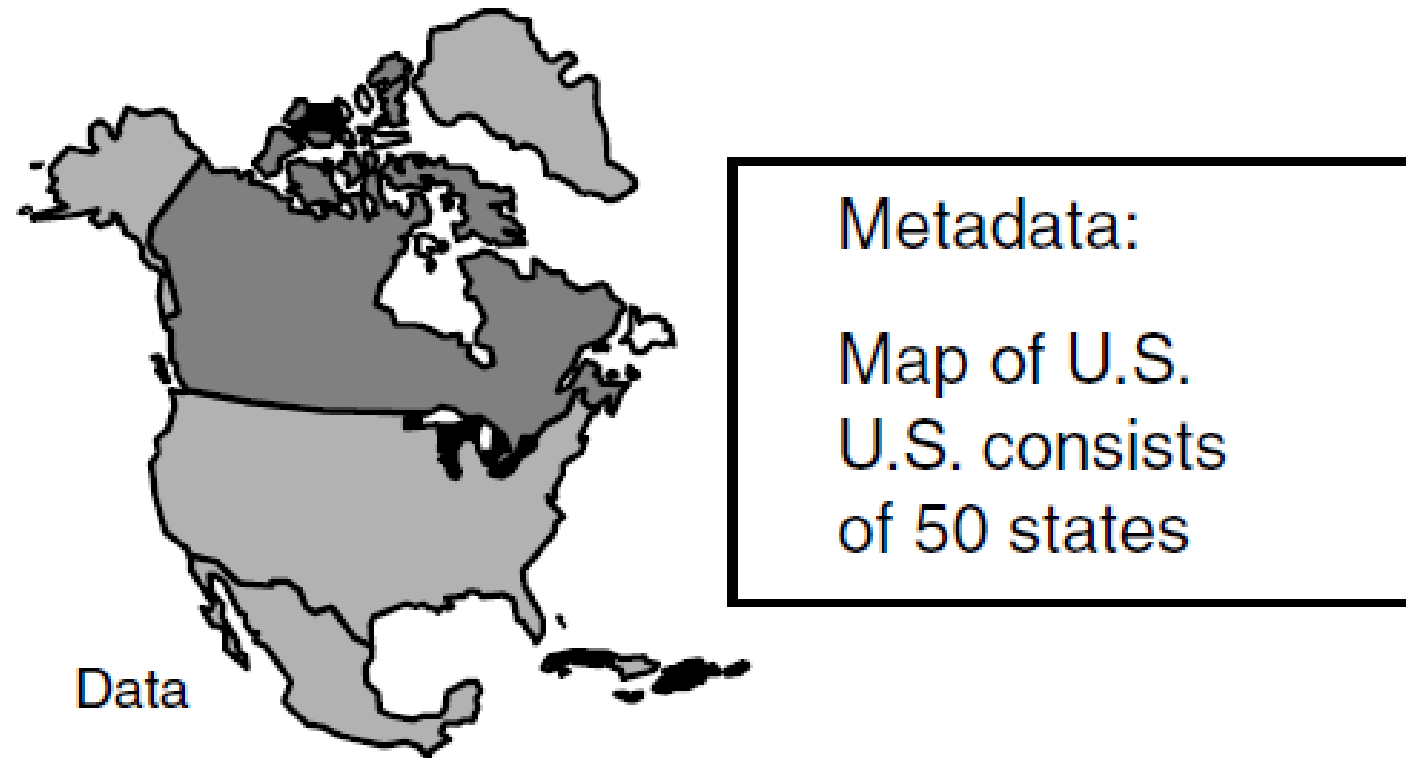
Image Metadata — Descriptive vs Content-Based

- ▶ Descriptive (semantic): natural-language caption, creator, location, event depicted
 - ▶ Example: 'Young woman with rose; yellow house and car in background'
- ▶ Technical/embedded: EXIF (camera, exposure, GPS), IPTC (caption, copyright, keywords)
- ▶ Content-based (CBIR): color histogram, texture (Gabor), shape, SIFT/ORB keypoints
- ▶ Relational model: Image table + ImageFeature table + Annotation table (1:N)
- ▶ Photos as metadata: thumbnail, alternate view, or proof image linked to primary record



Multimedia metadata

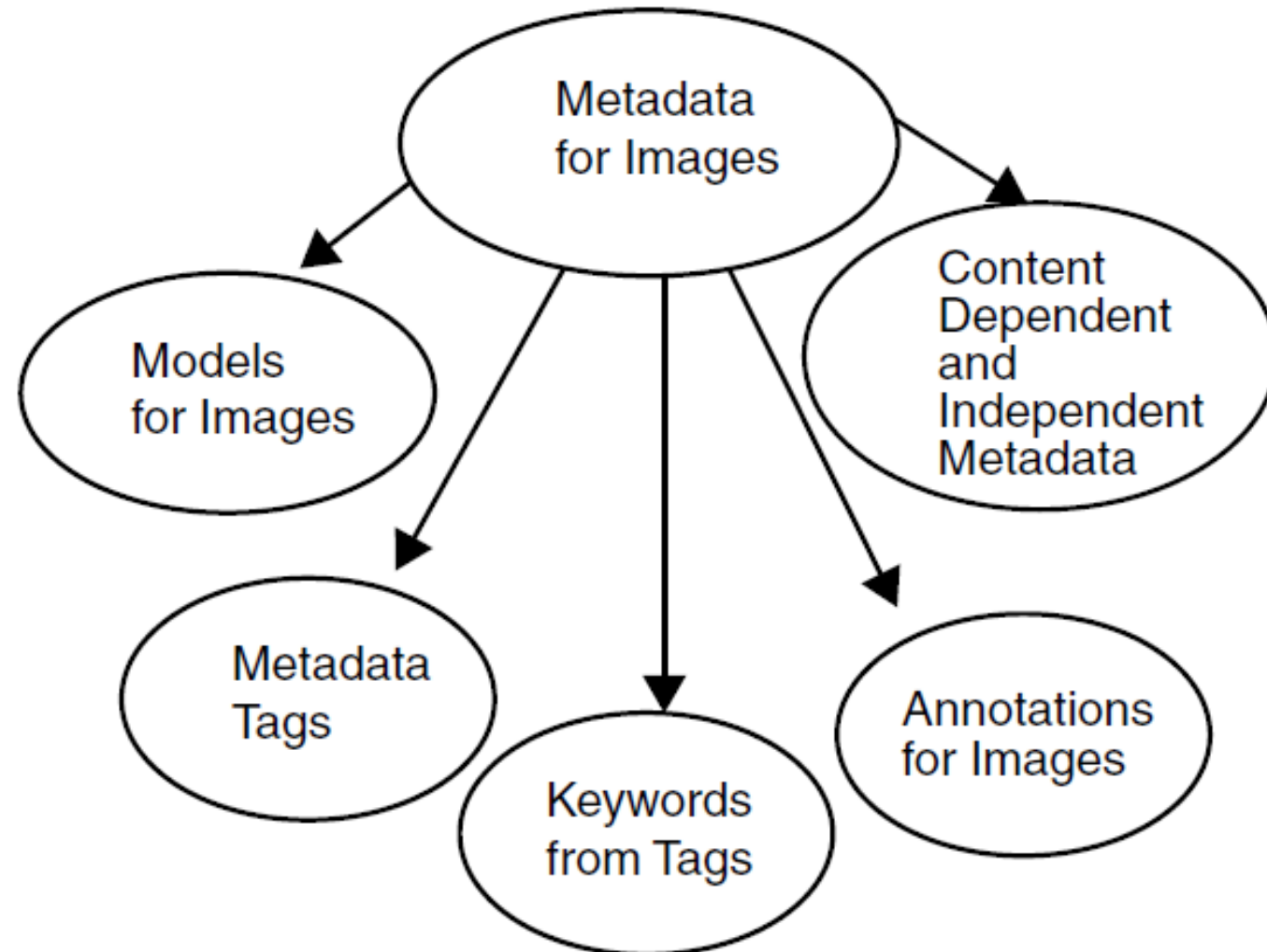
- ▶ Examples of image metadata



Example — Image Metadata Record

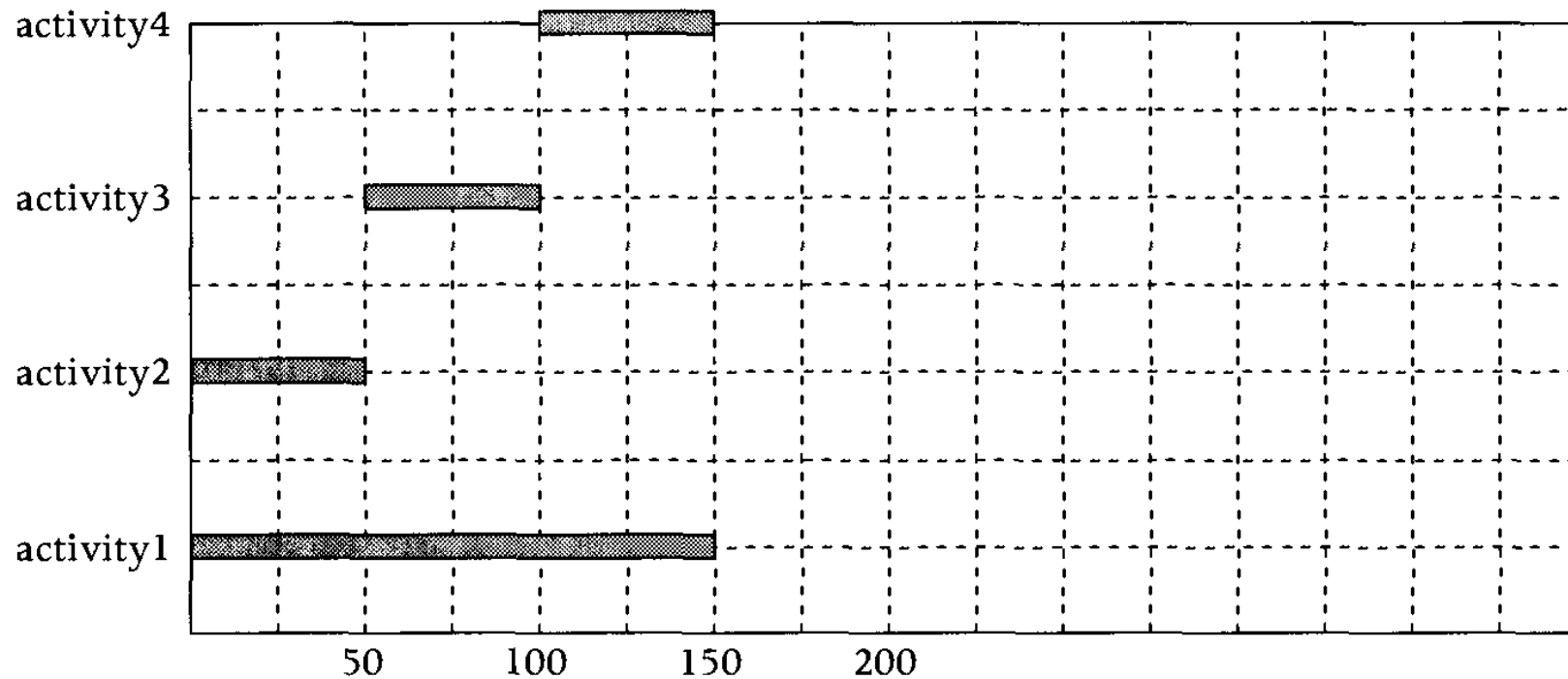
- ▶ Image ID: IMG_2024_001 | Format: JPEG | Resolution: 4032×3024 | Size: 2.1 MB
- ▶ EXIF: Camera=Canon EOS, ISO=400, Exposure=1/250s, GPS=(21.0285°N, 105.8542°E)
- ▶ IPTC Caption: 'Opening ceremony of Digital Heritage Exhibition, Hanoi'
- ▶ Keywords: {museum, exhibition, ceremony, heritage}
- ▶ CBIR features: dominant_colors={red:0.3, gold:0.4, blue:0.2}, edge_density=0.45
- ▶ Linked entity: Artwork OID b_exhibit_001 | Related audio: AUD_speech_001

Image metadata



Multimedia metadata

► Audio Metadata:



Audio Metadata — Structure and Standards

- ▶ Embedded tags: ID3v2 (MP3) — title, artist, album, genre, year, track, lyrics
- ▶ Temporal segmentation: audio as a sequence of labeled time intervals on a timeline
- ▶ Segment types: speech, music, silence, applause, background noise
- ▶ Per-segment attributes: speaker ID, language, transcript text, instrument, tempo
- ▶ MPEG-7 AudioDescription: timbre, tempo, melody contour, spoken content transcription
- ▶ Simultaneous layers: speech + music may overlap — metadata must model parallel streams



Multimedia metadata

- ▶ Metadata that represents audio content: each activity represents a specific content in the stored audio (e.g., singer singing, background sound, etc.). Each action may include different details:
 - ▶ voice
 - ▶ music
 - ▶ text
- ▶ Audio metadata includes multiple objects or content that occur simultaneously or consecutively in a time domain.

Multimedia metadata

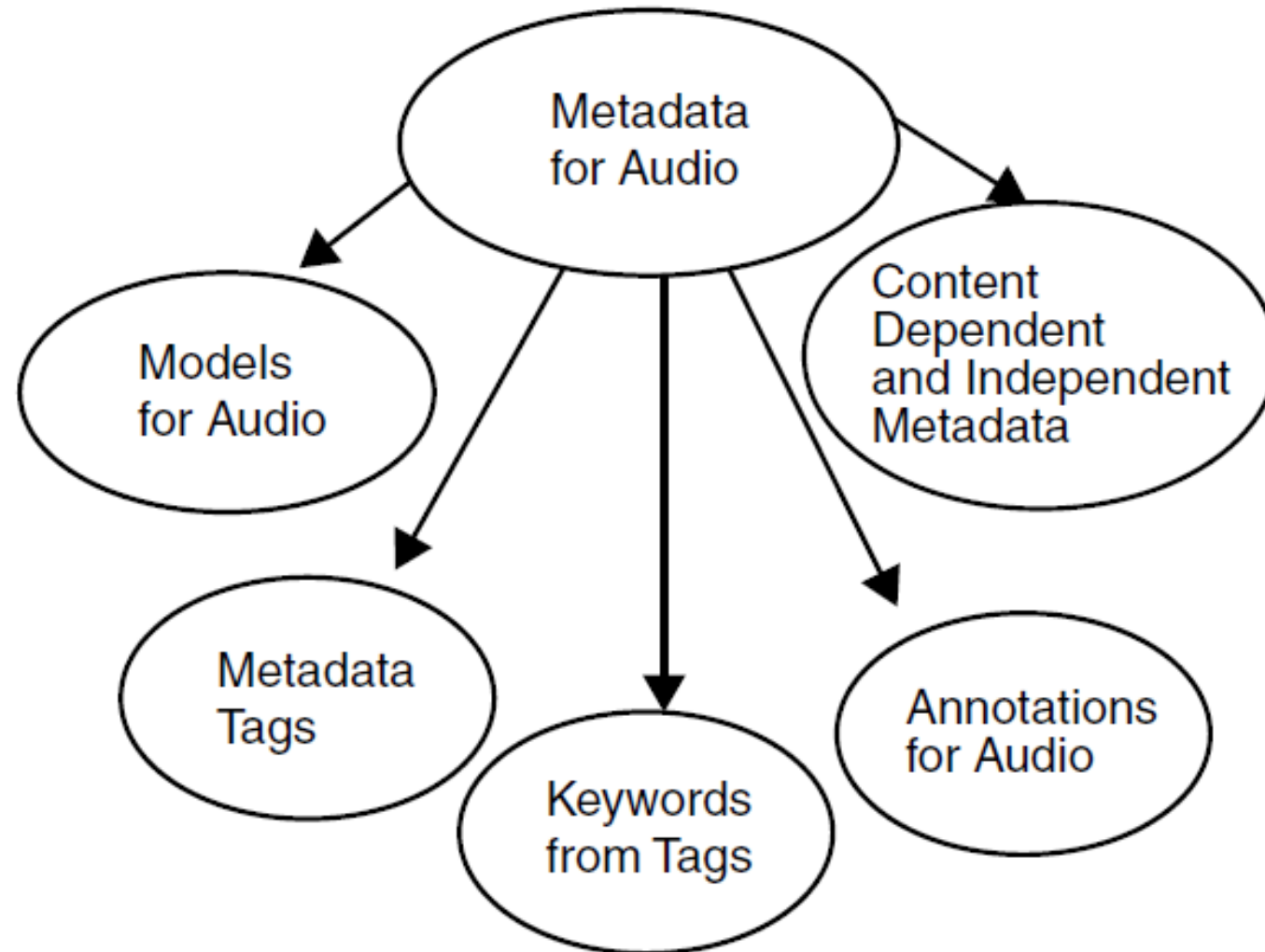
- ▶ For example, the metadata of an audio clip of a presidential speech with various content can be represented as follows:

Social Security Speech	Healthcare Speech	Foreign Policy Speech
Frames 1001-2000	Frames 2001-3000	Frames 3001-4000

Example — Audio Metadata Timeline

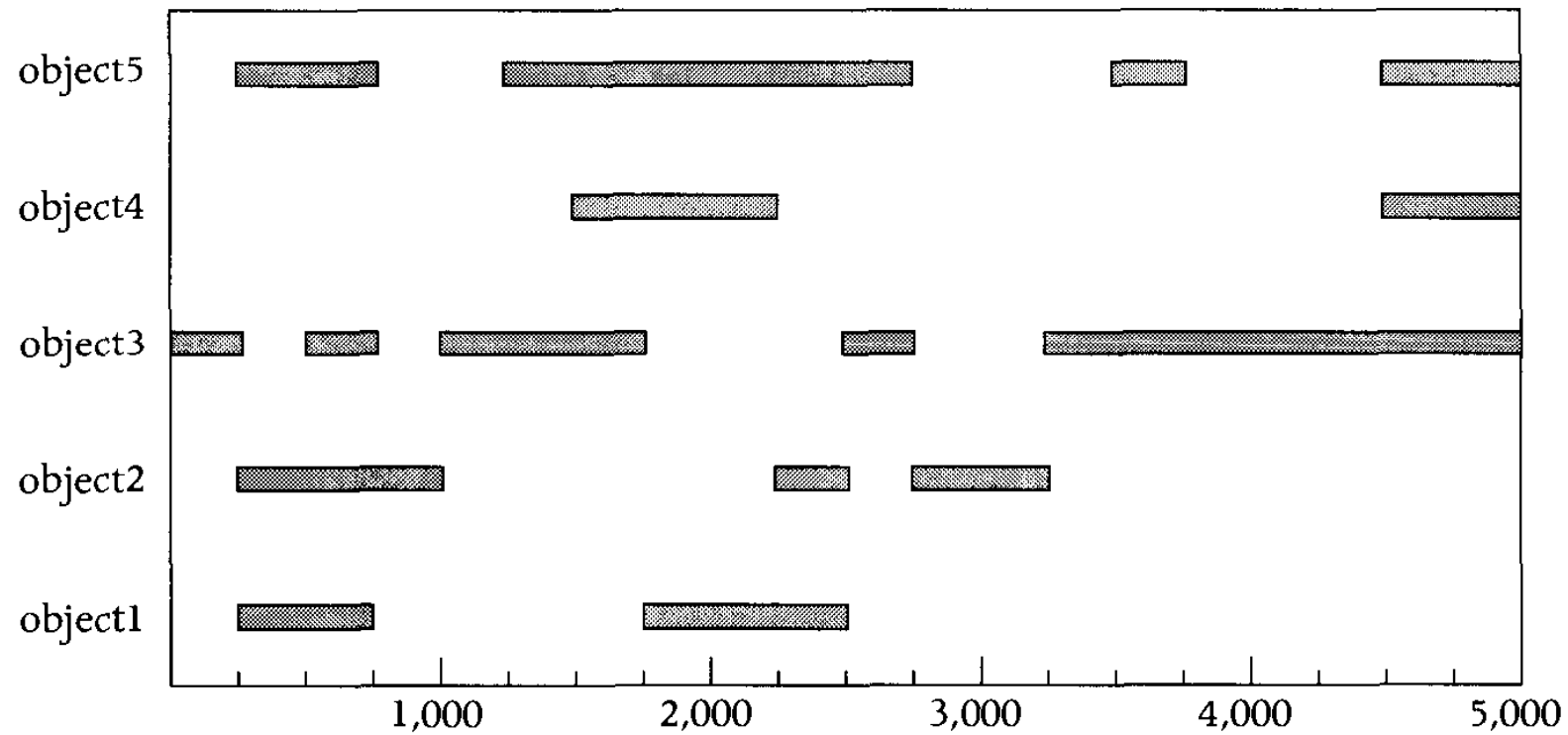
- ▶ Clip: Presidential speech (duration 4:30)
 - ▶ [0:00–0:15] type=music, content=opening_fanfare, volume=high
 - ▶ [0:15–1:45] type=speech, speaker=president, language=en, transcript=...
 - ▶ [1:45–1:50] type=applause, intensity=high
 - ▶ [1:50–3:30] type=speech, speaker=president, topic=economy
 - ▶ [3:30–4:30] type=music, content=closing_theme
- ▶ Global: format=MP3, bitrate=192kbps, sample_rate=44100Hz, channels=2

Audio metadata



Multimedia metadata

- ▶ Video Metadata: same as audio.

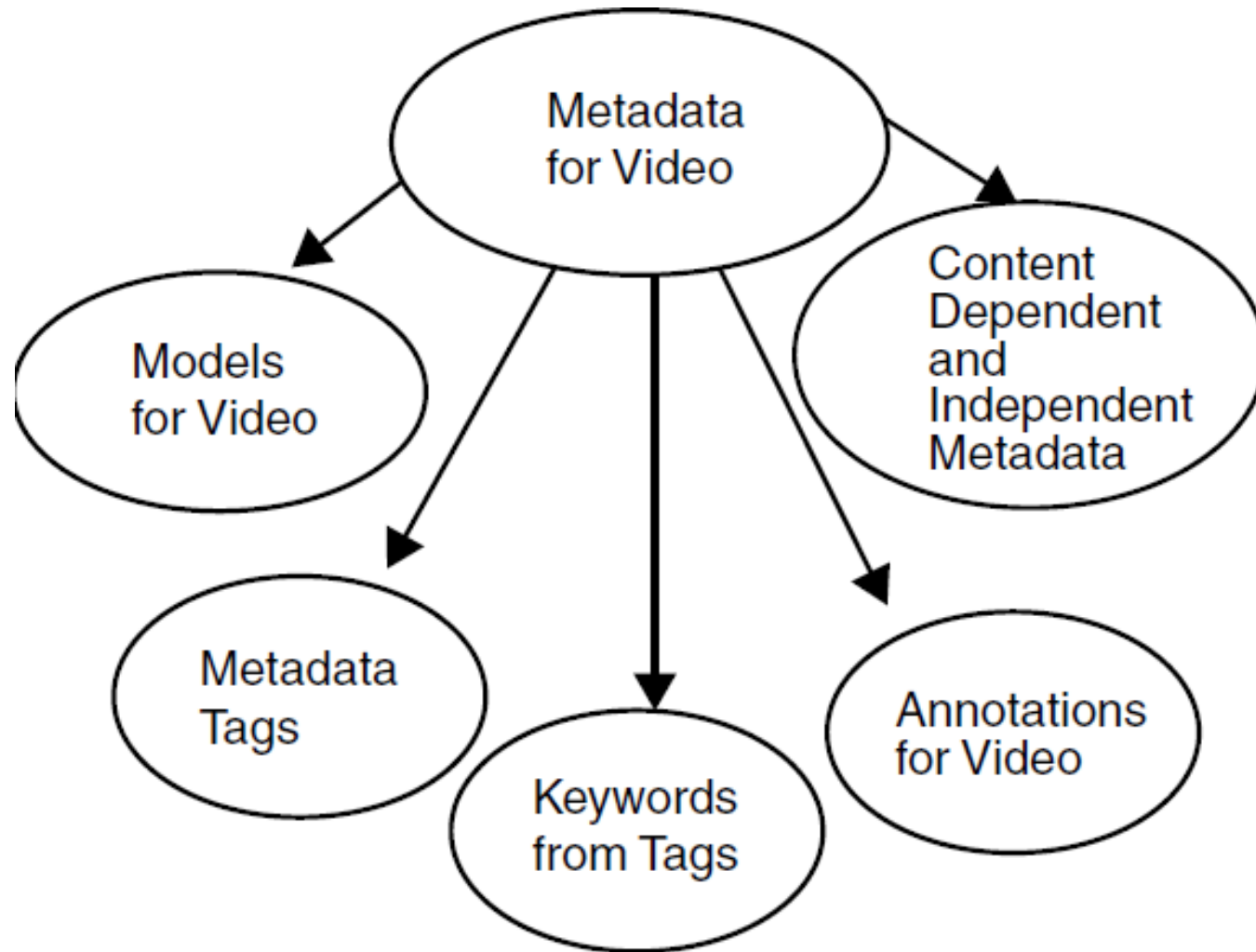


Video Metadata — Beyond Audio

- ▶ Inherits audio metadata plus spatial and visual temporal structure
- ▶ Keyframe metadata: representative frame per shot; thumbnail for browsing
- ▶ Scene/shot structure: scene boundaries, camera motion (pan, zoom, cut)
- ▶ Visual content per segment: objects detected, on-screen text (OCR), face IDs
- ▶ Synchronization: link subtitle track, slide deck, transcript to timeline offsets
- ▶ MPEG-7 VisualDescription + MediaTime — standard representation of spatiotemporal features



Video metadata

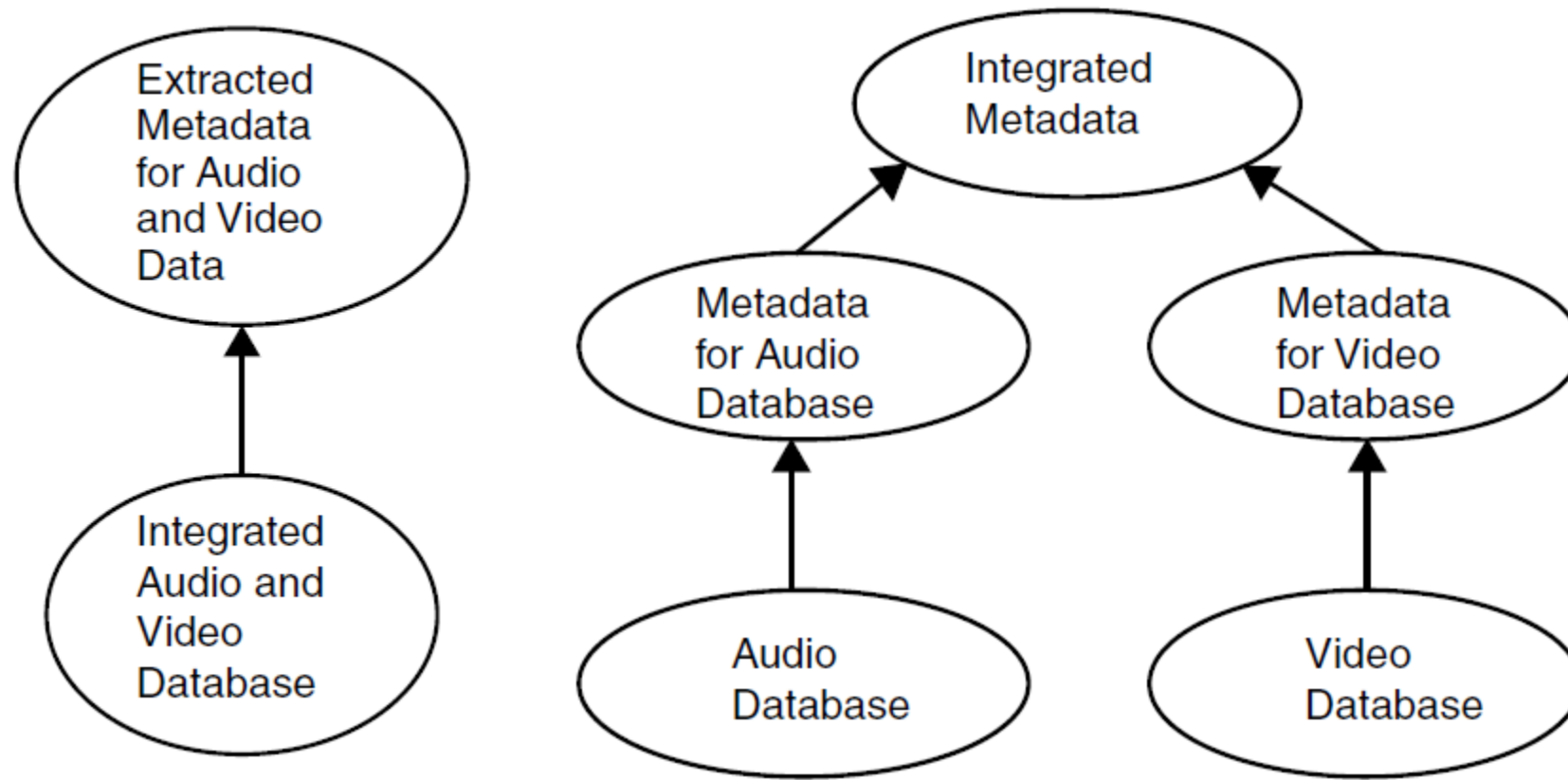


Example — Video Metadata Record

- ▶ Video: 'Lecture 9 Recording' | Duration: 45:00 | Format: MP4/H.264 | Resolution: 1080p
- ▶ Scenes:
 - ▶ Scene 1 [0:00–5:00]: title_slide, speaker=Dr.Nguyen, slide=1
 - ▶ Scene 2 [5:00–20:00]: presentation, topic=metadata_taxonomy, slides=2–10
 - ▶ Scene 3 [20:00–45:00]: Q&A, speaker=multiple
- ▶ Linked: transcript.txt, slides.pptx, audio_only.mp3
- ▶ Keyframes: kf_001@0:00, kf_002@5:00, kf_003@20:00

Multimedia metadata

- ▶ Metadata of audio and video integration data

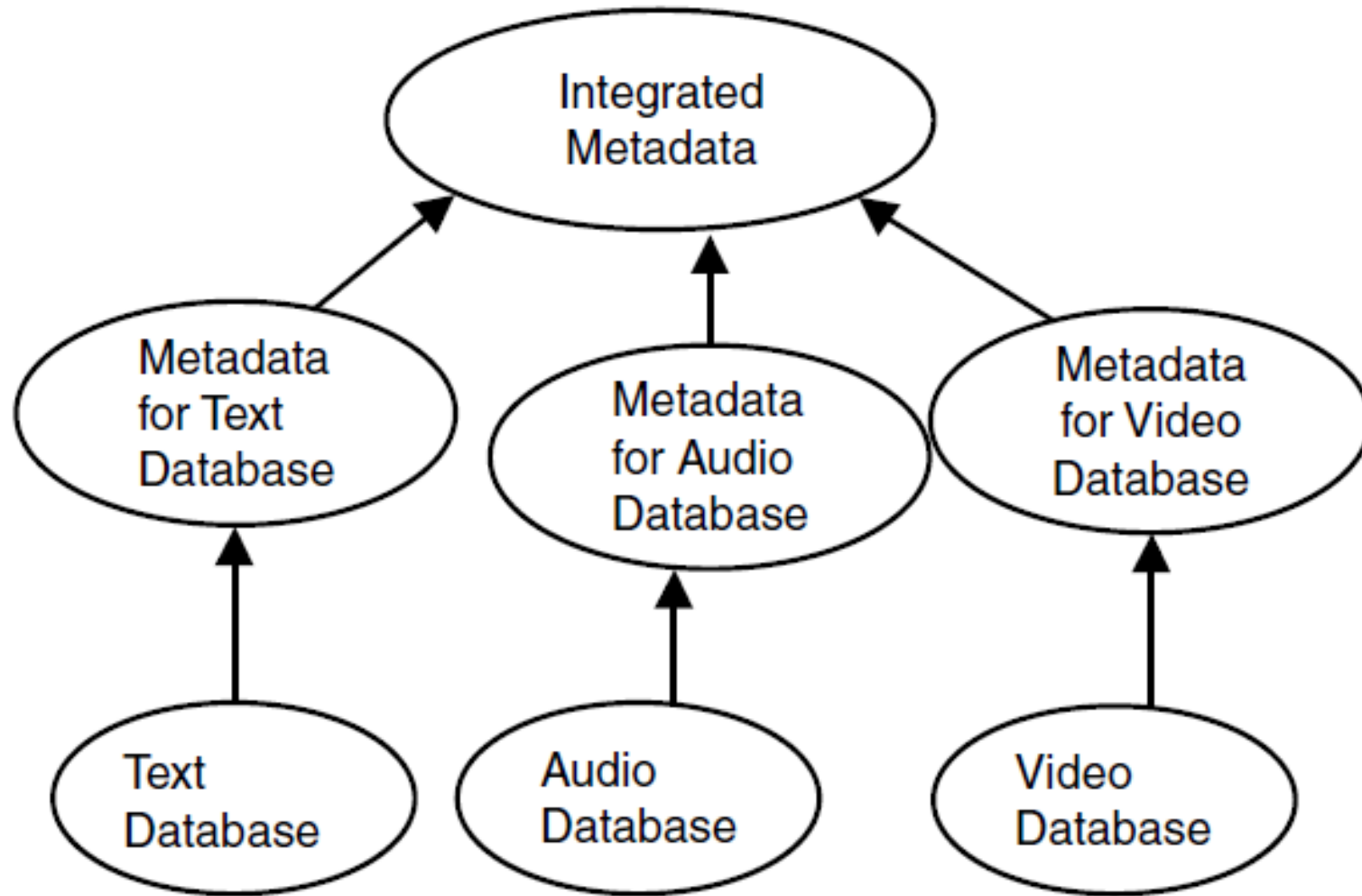


Integrated Audio-Video Metadata

- ▶ AV integration: single timeline unifying video track, audio track(s), subtitle, annotations
- ▶ Synchronization metadata: timecode origin, frame rate, audio sample offset, lip-sync tolerance
- ▶ Standards: SMIL (Synchronized Multimedia Integration Language), MXF (Material Exchange Format)
- ▶ Multi-track model: each track has type, codec, language, and segment list
- ▶ Cross-track links: subtitle line ↔ video frame ↔ transcript word (word-level alignment)
- ▶ Use case: museum kiosk — video tour + narration + hotspot coordinates on keyframes



Multimedia metadata



Example — Integrated AV Metadata

- ▶ Package: Museum_Tour_Room3.smil
 - ▶ Video track: tour_room3.mp4, 720p, 3:20 duration
 - ▶ Audio track 1: narrator_en.mp3 (English narration)
 - ▶ Audio track 2: ambient.mp3 (background)
 - ▶ Subtitle track: tour_room3.vtt (EN + VI)
 - ▶ Hotspots: [{time=1:10, x=320, y=240, target=artwork_OLD_42}, ...]
 - ▶ All tracks synchronized to t=0:00:00:00 @ 25fps

Multimedia metadata

- ▶ The concept of the three levels of expression(Panofsky)
 - ▶ Description
 - ▶ Identification
 - ▶ Interpretation

Three levels of expression (Panofsky)

- ▶ **Description level**
 - ▶ Representation of the entity's basic, natural information
 - ▶ The general components of the entity are depicted in works and images, ...
 - ▶ This information is generally known to any subject

Three levels of expression (Panofsky)

- ▶ Identification level

- ▶ The name of the entity can be contained in the database
- ▶ Capable of distinguishing different entities from each other
- ▶ The data can be aggregated of other data sources

Three levels of expression (Panofsky)

▶ Interpretation level

- ▶ Implications in the representation of entities in the database
- ▶ The meaning is often symbolized and often associated with the cultural framework where the database is formed
- ▶ Made by experts

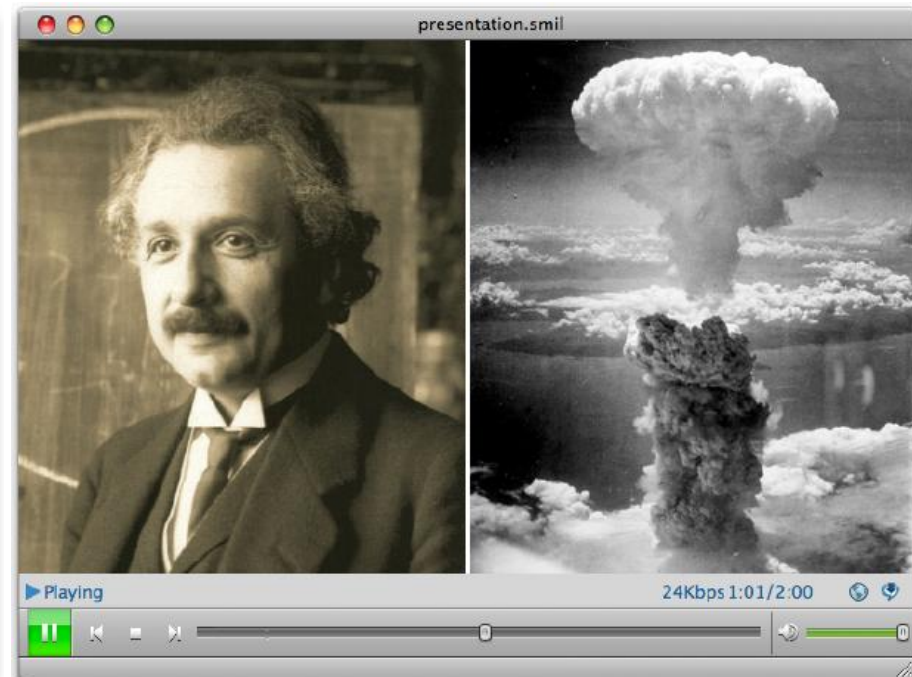
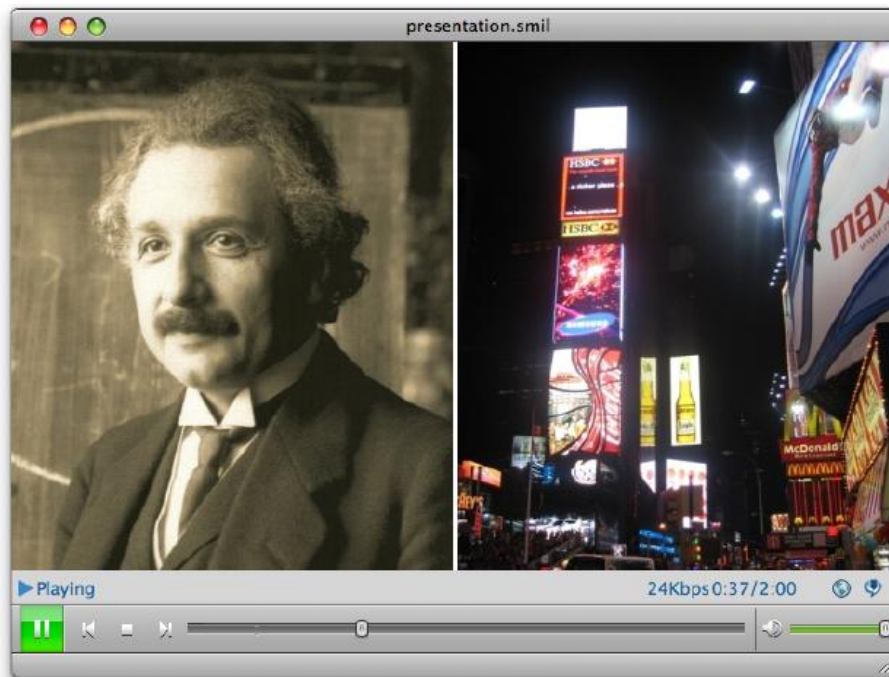
Panofsky Levels Mapped to Metadata Fields

- ▶ Description (pre-iconographic): raw perceptual facts — colors, shapes, objects, duration, format
 - ▶ Metadata: resolution, dominant colors, object labels ('person', 'building'), word count
- ▶ Identification (iconographic): who/what/when/where — names, dates, places, catalog IDs
 - ▶ Metadata: title, creator, creation date, location name, ISBN, museum accession number
- ▶ Interpretation (iconologic): symbolic/cultural meaning — themes, historical significance
 - ▶ Metadata: expert commentary, cultural context, symbolism, FRBR 'work' relationships
- ▶ Higher levels require more expert input; lower levels are more automatable



Example

- ▶ Albert Einstein and nuclear power.

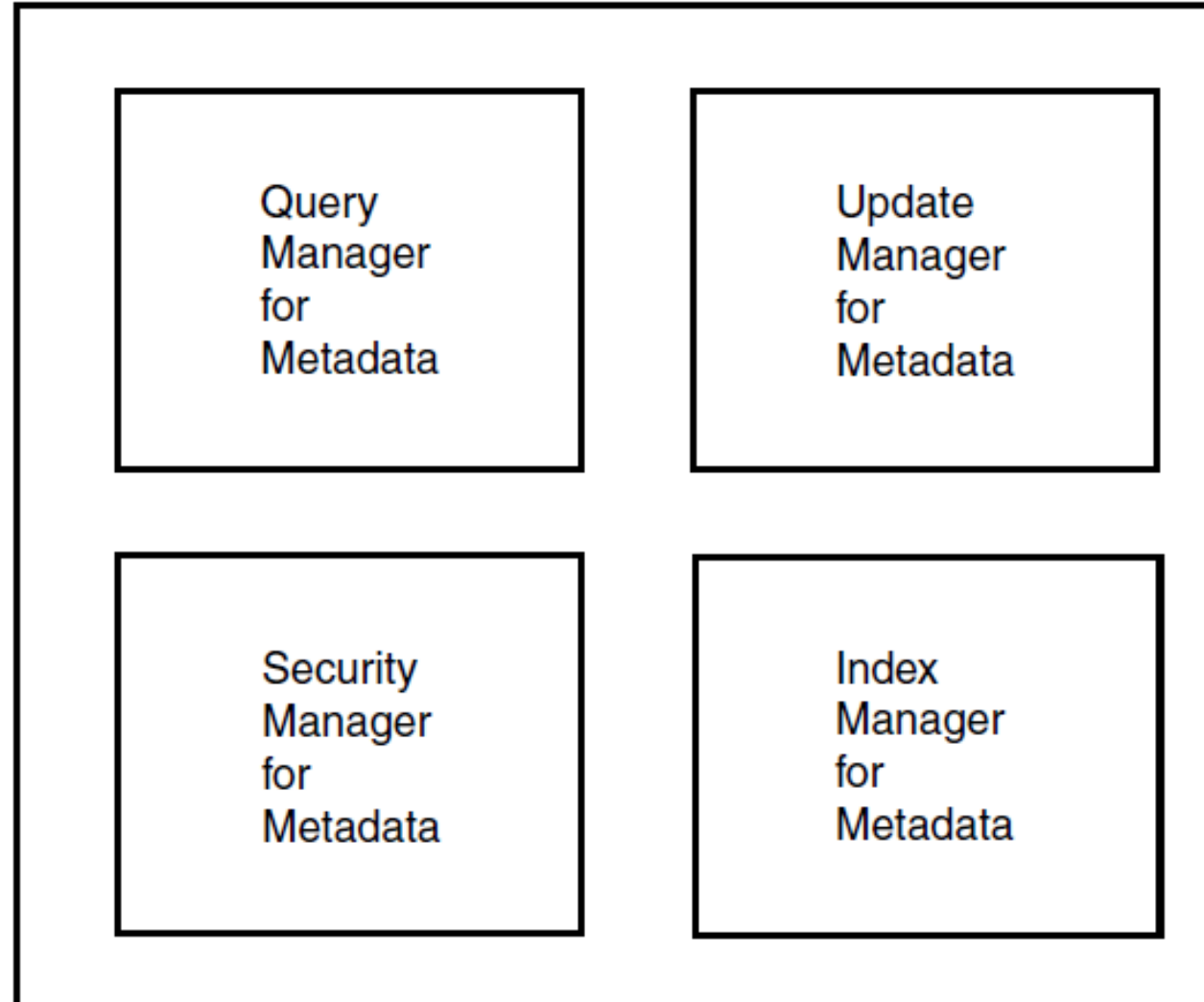


Example — Einstein Example Analysis

- ▶ For 'Albert Einstein and nuclear power', design metadata at all three Panofsky levels:
 - ▶ Description: list observable features in a photo/sculpture/document about Einstein
 - ▶ Identification: name, dates, field (physics), related events (Manhattan Project context)
 - ▶ Interpretation: cultural symbolism of Einstein–nuclear age; expert curator note
- ▶ Link 3 MM objects (portrait photo, documentary clip, encyclopedia text) via shared entity ID
- ▶ Which metadata fields should be shared across all 3 vs media-specific?



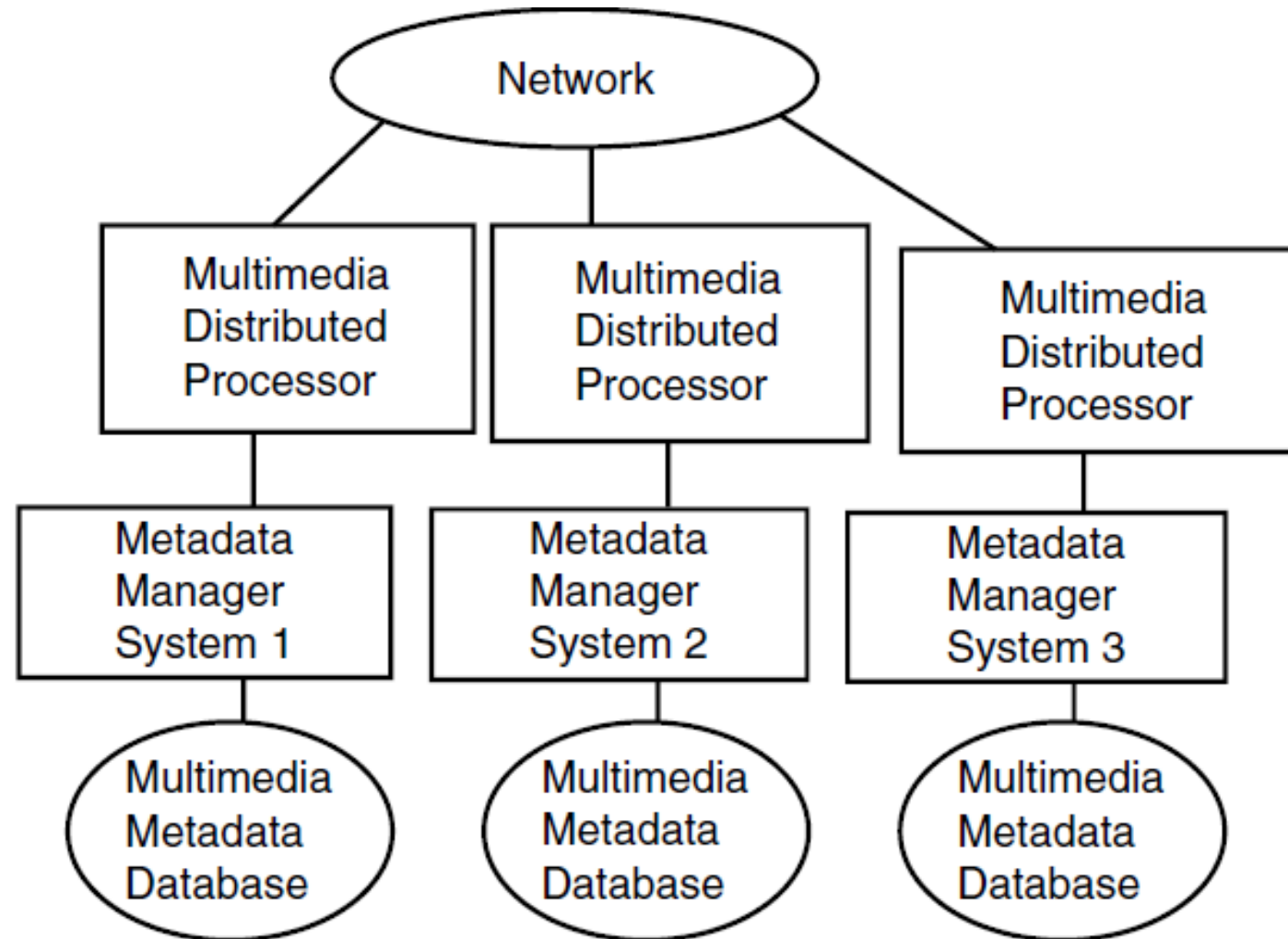
Multimedia metadata management



Multimedia Metadata Management Lifecycle

- ▶ Create: ingest media + extract/generate metadata at collection time
- ▶ Store: metadata repository (RDBMS, XML/JSON document store, triple store for linked data)
- ▶ Validate: schema compliance, controlled vocabulary checks, quality review workflow
- ▶ Index: catalog search (SQL), full-text (IR index), content-based (feature index)
- ▶ Update: version metadata when media is edited, migrated, or re-described
- ▶ Publish/Share: OAI-PMH harvest, API, linked open data (LOD) export

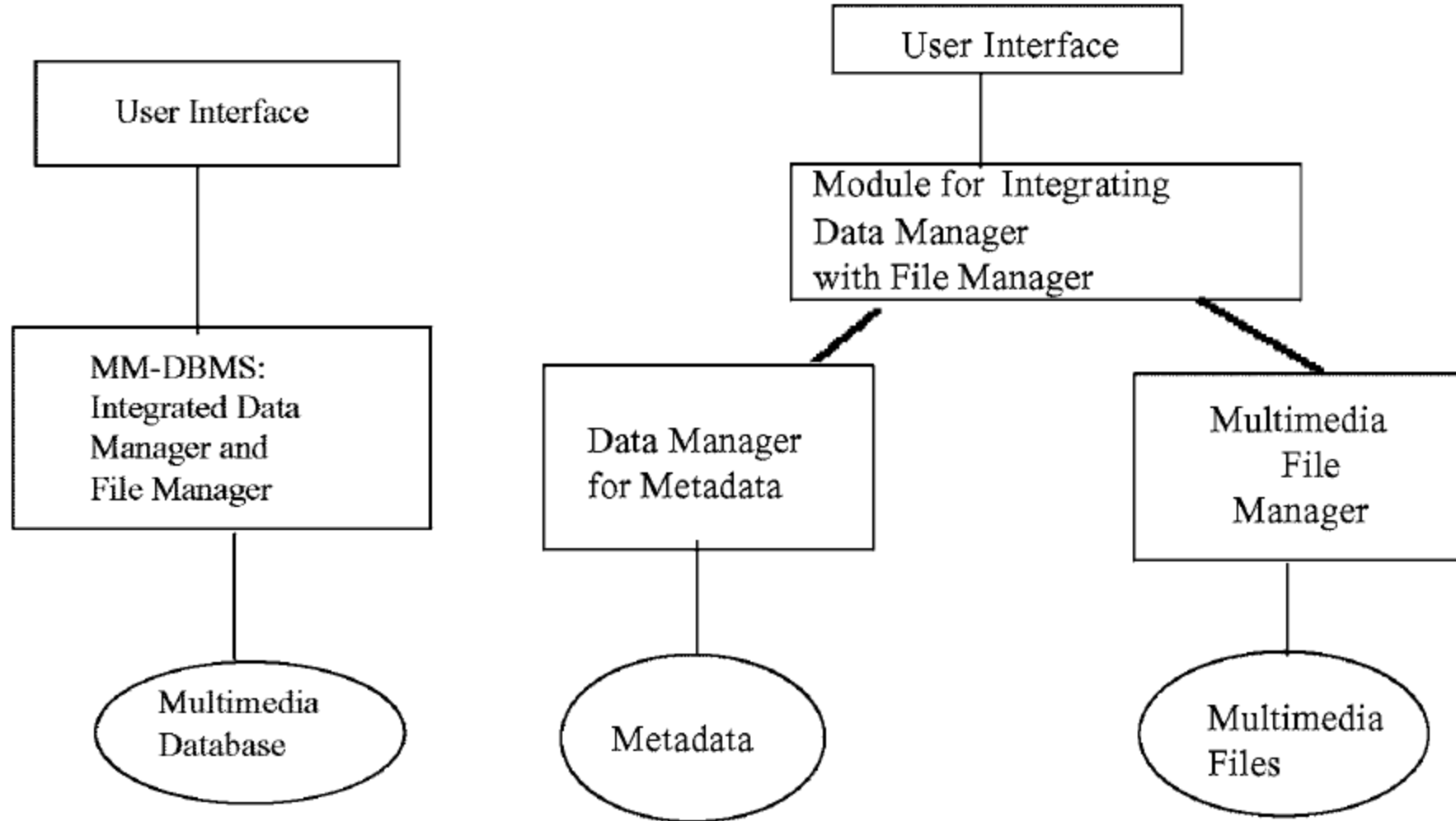
Distributed architecture for Multimedia metadata



Distributed Architecture for Metadata

- ▶ Central catalog + distributed media servers: metadata hub indexes OIDs; BLOBs stay at edge nodes
- ▶ Federated repositories: each museum/library holds local metadata; union catalog for cross-search
- ▶ Harvesting: OAI-PMH, CKAN — periodic metadata sync without moving media files
- ▶ Advantages: scalability, institutional autonomy, reduced bandwidth for large AV files
- ▶ Challenges: schema heterogeneity, ID reconciliation, consistency after local updates
- ▶ Solution patterns: shared ontology, crosswalk mappings (DC ↔ MODS ↔ MPEG-7)

Open architecture of Multimedia metadata



Open Architecture for Multimedia Metadata

- ▶ Open schema: extensible element set — institutions add domain-specific fields without breaking core
- ▶ Open APIs: REST/GraphQL metadata services; SPARQL for RDF/linked-data backends
- ▶ Open formats: XML (MODS, MPEG-7), JSON-LD, RDF/Turtle — avoid proprietary lock-in
- ▶ Plugin model: media-type handlers register extractors (ImageExtractor, AudioExtractor, ...)
- ▶ Benefits: interoperability, community extensions, long-term preservation
- ▶ Example stack: Dublin Core core + MPEG-7 media extensions + custom museum ontology

Controlled Vocabularies — Dictionary of Art

- ▶ Dictionary of Art (and similar): authoritative terms for art history, techniques, periods, artists
- ▶ Purpose: normalize identification metadata — 'oil on canvas' not 'oil paint on linen'
- ▶ Controlled vocabulary: preferred term + synonyms + broader/narrower term hierarchy
- ▶ Used at Identification and Interpretation levels (iconography, school, movement)
- ▶ Integration: metadata field 'style' must reference vocabulary ID, not free text
- ▶ Benefits: consistent search ('Renaissance' retrieves all mapped synonyms)

Dictionary of Art and Architecture

ID: 300011576

Record Type: [concept](#)

 **Carrara marble** (white marble, <marble by color or pattern>, ... Materials)

Note: Refers to a type of marble quarried in the area around Carrara, in Tuscany, Italy. It is characterized by a fine, compact grain and varies in color from pure white to creamy white, sometimes with a bluish tinge; it is a saccharoidal rock that can appear translucent in the finer grades. It has been a favorite stone of sculptors from antiquity to the present, including Michelangelo Buonarroti. Luna marble was the name used in ancient Rome.

Terms:

Carrara marble ([preferred](#), C,U,D,American English-P)

Luna marble (C,U,UF,American English)

marble, Carrara (C,U,UF,American English)

marble, Luna (C,U,UF,American English)

marmor Lunense (C,U,UF,American English)

marmor Luniense (C,U,UF,American English)

Facet/Hierarchy Code: [M.MT](#)

Hierarchical Position:

 [Materials Facet](#)
 [Materials](#)
 [materials](#)
 [<materials by composition>](#)
 [inorganic material](#)
 [rock](#)
 [metamorphic rock](#)
 [marble](#)
 [<marble by color or pattern>](#)
 [white marble](#)
 [Carrara marble](#)

Gazetteers — Dictionary of Place Names

- ▶ Gazetteer: authoritative register of place names with coordinates and hierarchical containment
- ▶ Fields: official name, alternate names, country, admin region, lat/long, place type
- ▶ Use in MM metadata: geo-tag photos/videos; normalize 'Saigon' vs 'Ho Chi Minh City'
- ▶ Links Identification-level metadata to maps and spatial search
- ▶ Standards: GeoNames, Getty TGN (Thesaurus of Geographic Names), ISO 19112
- ▶ Cross-media: same place ID links landscape photo, field recording, and travel article

Dictionary of place names

ID: 7006077

Record Type: **administrative**

 **Carrara (inhabited place)**

Coordinates:

Lat: 44 05 00 N *degrees minutes* Lat: 44.0833 *decimal degrees*
Long: 010 06 00 E *degrees minutes* Long: 10.1000 *decimal degrees*

Note: Located on the Carrione river; gained importance as marble-quarrying center, noted for fine white marble suitable for sculpture; was important medieval center; was center of Massa-Carrara principality & duchy (after 1633).

Names:

Carrara (**preferred**, C,V,N)

Apuania (H,V,N) ancient name

Hierarchical Position:

 World (facet)
 Europe (continent)
 Italy (nation)
 Tuscany (region) (C) part of new kingdom of Italy from 1861
 Massa-Carrara (province)
 Carrara (inhabited place)

Place Types:

inhabited place (**preferred**, C) was an ancient center
city (C)
quarrying center (C)
episcopal see (C)

Summary — Multimedia Metadata

- ▶ MM data differs from human perception — metadata bridges the expression and semantic gaps
- ▶ Four metadata types: descriptive, structural, administrative, technical
- ▶ Each media type adds specialized metadata: text catalog, EXIF/CBIR, audio timeline, video scenes
- ▶ Panofsky's three levels guide how deeply metadata captures meaning
- ▶ Standards (DC, MPEG-7, MODS) and controlled vocabularies enable interoperability

